

v38 Addendum: Field Initialization and Absorbing Boundary Conditions Revisited

Linearized GR + EM PIC Sandbox, web-frontend phase

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Scope. This addendum revises three sections of `gr_sandbox_v38.tex` that, while internally consistent, were found to be physically wrong or numerically broken in the course of building the interactive web demo against the v40/v41 simulator core:

1. §9.6 “Absorbing boundary layer”: the multiplicative (σdt) damping ring does not absorb the lowest standing mode of the bounded box. Replace with a derivative-friction (Klein-Gordon-like) absorber.
2. §15.9 “Field initialization”: the wave-equation convergence iteration starting from $\Phi = 0$ does not converge to a static solution for the same reason – the lowest standing mode, seeded by the initial transient, is undamped. Replace with a direct-sum 2D Liénard-Wiechert initialization that sets all six potentials analytically.
3. Add a new subsection on the DC gauge mode: under Neumann outer BC the constant-everywhere mode of a scalar potential is accelerated by any nonzero source mean and runs away; under Dirichlet the wall pins it. This was implicit in v38 but the failure mode wasn’t called out.

The discrete stability analysis of the centered-implicit friction form (which the post-step “bleed” form is *not*) is included because the result is non-trivial and easy to get wrong.

Contents

1 Diagnosis of the v38 absorber

The v38 absorbing ring multiplies the leapfrog update by $(1 - \sigma_k dt)$ in a polynomial-tapered ring of depth N cells:

$$\Phi_k^{n+1} = [2\Phi_k^n - \Phi_k^{n-1} + c^2 dt^2 (\nabla^2 \Phi^n + s_c \rho_k)] (1 - \sigma_k dt). \quad (1)$$

This factor does two distinct things: (a) it dissipates outgoing wave energy, which is what we want; and (b) it pulls Φ toward zero everywhere inside the ring, which effectively imposes a Dirichlet boundary at the wall. Effect (a) only acts on field amplitude that *reaches* the ring; effect (b) is the reason the static fixed point of the box is the Dirichlet Poisson solution rather than the free-space one.

The failure mode is mode-dependent. Consider the standing modes $\phi_{mn}(x, y) = \sin(m\pi x/L) \sin(n\pi y/L)$ of a square box with zero-Dirichlet walls. The lowest mode $(m, n) = (1, 1)$ has its **antinode at the center** and **nodes at the wall**. Its overlap with the multiplicative damping array, concentrated in the outer ring, is suppressed by the eigenfunction’s small wall amplitude. Specifically, the per-step decay factor for mode ϕ_{mn} is approximately

$$1 - dt \bar{\sigma}_{mn}, \quad \bar{\sigma}_{mn} = \frac{\int_{\Omega} \sigma(x, y) \phi_{mn}^2 d^2x}{\int_{\Omega} \phi_{mn}^2 d^2x}. \quad (2)$$

For the lowest mode this average is $\mathcal{O}(N/L)$ times the wall value of σ ; for high- k modes (which oscillate with appreciable amplitude at the wall) it is $\mathcal{O}(1)$ times the wall value. At $N/L \sim 1/8$ (a typical absorber ring depth) the lowest mode is damped roughly an order of magnitude slower than high- k modes.

Empirical signature. With `pic.static` (a single stationary particle), particles frozen, and the field initialized by the v38 §15.9 convergence iteration, the value of Φ_g at the box center oscillates with period $\sim 2L/c$ and amplitude $\sim |\Phi_g^{\text{static}}|$ indefinitely. The probe trajectory shows envelope swing of $\sim 50\%$ of $|\Phi_g^{\text{static}}|$ after 4000 steps with no decay trend. The absorbing ring is not absorbing the lowest mode.

The v38 §15.9 prescription – “iterate the leapfrog with frozen particles until the field converges” – relies on the absorber to remove the transient launched by switching the source on at $t = 0$. That transient has Fourier content at every mode; the lowest component is excited as strongly as any other, and once excited it persists. The convergence iteration therefore does not, in fact, converge to the static solution; it converges to the static solution *plus* a residual standing-mode oscillation of comparable amplitude.

2 Direct-sum Liénard-Wiechert initialization

The correct initialization is to set each perturbation potential at $t = 0$ to the static Poisson solution of the deposited sources, then start the wave equation from there with zero time-derivative. For an isolated N -particle configuration in 2D, the static solution is known analytically – it is the 2D Liénard-Wiechert potential, which under the Lorenz gauge is built from the same $\ln r$ kernel for all six fields:

$$\Phi_g(\mathbf{x}) = -\frac{s_c^g}{2\pi} \sum_i m_i \ln |\mathbf{x} - \mathbf{x}_i|, \quad (3)$$

$$\mathbf{A}_g(\mathbf{x}) = -\frac{s_c^g}{2\pi c^2} \sum_i m_i \mathbf{v}_i \ln |\mathbf{x} - \mathbf{x}_i|, \quad (4)$$

$$\phi(\mathbf{x}) = -\frac{s_c^{em}}{2\pi} \sum_i q_i \ln |\mathbf{x} - \mathbf{x}_i|, \quad (5)$$

$$\mathbf{A}(\mathbf{x}) = -\frac{s_c^{em}}{2\pi c^2} \sum_i q_i \mathbf{v}_i \ln |\mathbf{x} - \mathbf{x}_i|, \quad (6)$$

where $s_c^g = -4\pi G_{\text{eff}}$ and $s_c^{em} = +4\pi k_e$ are the source coefficients from §9.2, and $\mathbf{v}_i = \mathbf{p}_i/(\gamma_i m_i)$. These follow from the 2D fundamental solution of ∇^2 , $G(r) = (2\pi)^{-1} \ln r$, applied to point-source charge/mass and current distributions $\rho(\mathbf{y}) = \sum_i Q_i \delta(\mathbf{y} - \mathbf{x}_i)$. The velocity moves from the scalar to the vector exactly as in the 3D Liénard-Wiechert form; in the static-source limit the scalar parts dominate and the vector parts are an $\mathcal{O}(v/c)$ correction.

Sub-cell clip

The logarithm diverges at $\mathbf{x} = \mathbf{x}_i$. Clip the radius at $r_{\min} = \frac{1}{2} dx$ (smaller than any cell-particle separation that can actually occur on the half-cell-centered deposition lattice) when evaluating the analytic sum; the resulting value at the source cell is finite and within the floating-point dynamic range of all nearby cells. The clipped value is not the discrete 5-point Poisson solution – see §?? – but the residual mismatch is handled by the friction absorber introduced below.

Boundary taper

If the outer wall enforces zero-Dirichlet (the v38 convention, kept here for reasons discussed in §??), the L-W field at the wall is nonzero and inconsistent with the BC. Multiply the L-W field

through the absorbing zone by a smooth taper that brings it to zero at the wall:

$$\Phi(\mathbf{x}) \leftarrow T(d(\mathbf{x})) \Phi(\mathbf{x}), \quad T(d) = \begin{cases} 1 & d \geq N_{\text{damp}} \\ 1 - u^2, \quad u = (N_{\text{damp}} - d)/N_{\text{damp}} & 0 < d < N_{\text{damp}} \\ 0 & d \leq 0 \end{cases} \quad (7)$$

where $d(\mathbf{x})$ is the cell-count distance from the nearest wall. This matches the polynomial $m = 2$ damping profile of §9.6. Under Neumann BC the taper must be *omitted*; see §??.

Leapfrog initialization at $t = 0$

After the L-W direct sum is written into the curr buffer of each field, also write it into the prev and next buffers:

$$\Phi_k^{-1} = \Phi_k^0 = \Phi_k^1 = \Phi_k^{\text{LW}}. \quad (8)$$

This makes the discrete time derivative encoded by the leapfrog ($\Phi^0 - \Phi^{-1}$) exactly zero, so no spurious transient is launched by the time-integration startup. Particle momenta are set directly from the scenario initial conditions; the v38 §15.9 half-step back-extrapolation is not needed because the field is already at the static solution.

Compute cost

For N_p particles on a $W \times H$ grid the L-W sum is $\mathcal{O}(N_p \cdot W \cdot H)$ log evaluations. At $W = H = 384$, $N_p = 2$ this is $\sim 3 \times 10^5$ operations – sub-millisecond. For $N_p \gtrsim 10^4$ a tree-code or FFT convolution would be needed, but the regime the sandbox addresses (a few orbiting test masses) does not require it.

3 Continuum–discrete Poisson mismatch

The L-W direct sum is the analytic Poisson solution of a sum of Dirac delta sources. The simulation’s actual source is not a delta; it is the TSC-spread (or BUMP, or CIC) deposition kernel evaluated at the particle’s sub-cell position. Inserting the analytic L-W into the discrete 5-point leapfrog and evaluating the residual $\nabla_5^2 \Phi^{\text{LW}} + s_c \rho^{\text{TSC}}$ shows that the two are not equal at the source. At a single particle of mass m centered exactly on a cell, with TSC weights (9/16, 3/32, 1/64) on the 3×3 stamp:

$$\nabla_5^2 \Phi^{\text{LW}}|_{\text{center}} = -\frac{4 \Phi_{\text{center}}^{\text{LW}}}{dx^2} \approx +5.55 \times 10^{-2} \quad (\text{at } m = 0.01, G_{\text{eff}} = 1), \quad (9)$$

while

$$s_c \rho_{\text{center}}^{\text{TSC}} = -4\pi G_{\text{eff}} \frac{m \cdot 9/16}{dx^2} \approx -7.07 \times 10^{-2}. \quad (10)$$

The residual -1.5×10^{-2} is roughly 20% of $|s_c \rho|$ at the source cell. It produces a per-step kick on Φ_g at the source of $c^2 dt^2 \times (-1.5 \times 10^{-2}) \approx -8 \times 10^{-3} \sim 5\%$ of $|\Phi_g^{\text{LW}}|_{\text{center}}$ every step. This is the source-term forcing that the friction absorber (next section) must dissipate.

Why not solve the discrete 5-point Poisson exactly? An SOR sweep on $\nabla_5^2 \Phi = -s_c \rho^{\text{TSC}}$ converges in $\mathcal{O}(\max(W, H))$ iterations to the bit-exact discrete Poisson solution and removes this residual entirely. This was implemented and tested. However, the SOR solution requires a boundary condition: zero-Dirichlet at the outer ring is the natural choice, but the resulting static field is the Dirichlet-Poisson solution of the box, not the free-space 2D log. The two differ visibly near the wall, where the Dirichlet enforcement forces $\Phi \rightarrow 0$ rather than continuing the $\ln r$ tail. For the pedagogical sandbox the free-space appearance of L-W is preferable, so we accept the small discrete-vs-continuum mismatch and rely on the friction absorber to dissipate the resulting per-step forcing.

4 Derivative-friction absorber

Replace the multiplicative damping ring of §9.6 with a derivative- friction term added to the wave equation:

$$\frac{\partial^2 \Phi}{\partial t^2} + 2\gamma(\mathbf{x}) \frac{\partial \Phi}{\partial t} = c^2 \nabla^2 \Phi + s_c \rho. \quad (11)$$

The crucial difference from the multiplicative form is that the friction term vanishes identically at $\partial_t \Phi = 0$, so the static fixed point is exactly the Poisson solution regardless of the spatial profile of γ . In particular: a uniform friction floor γ_0 everywhere in the box damps all modes – including the lowest standing mode, which has nonzero $\partial_t \Phi$ even though its spatial amplitude at the wall is small – without shifting the equilibrium.

4.1 Mode-by-mode decay rate

Consider a single eigenmode $\Phi_k(\mathbf{x}, t) = a_k(t) \phi_k(\mathbf{x})$ of the spatial operator $-c^2 \nabla^2$ with eigenvalue ω_k^2 , and let $2\bar{\gamma}_k$ denote the eigenmode-averaged friction. In the source-free regime:

$$\ddot{a}_k + 2\bar{\gamma}_k \dot{a}_k + \omega_k^2 a_k = 0 \implies a_k(t) \propto e^{-\bar{\gamma}_k t} \cos(\sqrt{\omega_k^2 - \bar{\gamma}_k^2} t + \delta). \quad (12)$$

For $\bar{\gamma}_k < \omega_k$ (the underdamped regime) every mode decays at rate $\bar{\gamma}_k$. The lowest mode’s eigenmode-averaged friction is $\bar{\gamma}_{(1,1)} \sim \gamma_0$ (dominated by the uniform floor, since the wall ramp contributes only at the small-amplitude boundary). A uniform $\gamma_0 dt = 10^{-3}$ gives mode-amplitude decay to $1/e$ in ~ 1000 steps. Higher- k modes are also damped by the wall ramp; combined decay rates approach γ_{wall} for modes localized near the boundary.

4.2 Discrete leapfrog: post-step bleed (UNSTABLE)

The natural-looking discretization of (??) is to run the standard leapfrog and then, after each rotation, bleed the prev buffer toward curr:

$$\Phi_{k,\text{bled}}^{n-1} := (1 - \alpha_k) \Phi_k^{n-1} + \alpha_k \Phi_k^n, \quad \alpha_k = 2\gamma_k dt. \quad (13)$$

This appears to reduce the encoded time derivative $\Phi^n - \Phi^{n-1}$ by a factor $(1 - \alpha_k)$ each step without touching the static fixed point (where $\Phi^n = \Phi^{n-1}$ and the bleed is a no-op).

This form is unconditionally unstable at the 2D Nyquist mode for any $\alpha > 0$ when CFL = $1/\sqrt{2}$.

The recurrence implied by (??) is

$$\Phi^{n+2} = (2 - \alpha - s^2) \Phi^{n+1} - (1 - \alpha) \Phi^n + c^2 dt^2 \cdot (\text{source terms}), \quad (14)$$

where $s^2 = c^2 dt^2 |\nabla_5^2|$. At the 2D Nyquist ($s^2 = 4$ when CFL = $1/\sqrt{2}$), the characteristic equation $\lambda^2 - (2 - \alpha - s^2)\lambda + (1 - \alpha) = 0$ has discriminant $\alpha^2 + 8\alpha > 0$ for any $\alpha > 0$. The roots are

$$\lambda_{\pm} = \frac{-2 - \alpha \pm \sqrt{\alpha^2 + 8\alpha}}{2}. \quad (15)$$

For $\alpha = 10^{-3}$, $\lambda_- \approx -1.045$ – the Nyquist mode grows 4.5% per step. Round-off seeds it; saturation in ~ 100 steps.

Empirically (the failure mode that prompted this addendum): on a 384×384 grid with L-W initialization and post-step-bleed friction at $\alpha = 0.04$ near the wall, a coherent wavefront appears at the boundary within a few hundred steps, propagates inward, and dominates the entire field within a few thousand. The “absorbing layer” is acting as a generator.

4.3 Centered-implicit discretization (STABLE)

Discretize the friction term at the leapfrog midpoint, so the friction gradient term $\partial_t \Phi$ uses $(\Phi^{n+1} - \Phi^{n-1})/(2dt)$:

$$\frac{\Phi^{n+1} - 2\Phi^n + \Phi^{n-1}}{dt^2} + 2\gamma \frac{\Phi^{n+1} - \Phi^{n-1}}{2dt} = c^2 \nabla_5^2 \Phi^n + s_c \rho^n. \quad (16)$$

Solving for Φ^{n+1} with $\beta = \gamma dt$:

$$(1 + \beta) \Phi^{n+1} = 2\Phi^n - (1 - \beta) \Phi^{n-1} + dt^2 (c^2 \nabla_5^2 \Phi^n + s_c \rho^n). \quad (17)$$

At the static fixed point $\Phi^{n+1} = \Phi^n = \Phi^{n-1}$ the LHS factor $(1 + \beta)$ and the RHS coefficient $-(1 - \beta) + 2 = 1 + \beta$ match identically – the static is preserved.

Stability at 2D Nyquist. The characteristic equation

$$(1 + \beta)\lambda^2 - (2 - s^2)\lambda + (1 - \beta) = 0 \quad (18)$$

at $s^2 = 4$ has discriminant $(2 - 4)^2 - 4(1 - \beta)(1 + \beta) = 4\beta^2$, giving real roots

$$\lambda_+ = -\frac{1 - \beta}{1 + \beta}, \quad \lambda_- = -1. \quad (19)$$

$|\lambda_+| < 1$ for $\beta > 0$; $|\lambda_-| = 1$. The Nyquist mode is marginally stable on one branch and damped on the other – unconditionally non-growing for any $\beta \geq 0$. At lower- k modes the discriminant is negative and $|\lambda|^2 = (1 - \beta)/(1 + \beta) < 1$ for $\beta > 0$: every interior mode strictly decays.

Post-leapfrog correction implementation. Equation (??) is equivalent to running the standard (un-friction) leapfrog and then applying a per-cell post-correction *before* the buffer rotation:

$$\Phi_k^{n+1} \leftarrow \frac{\Phi_{k,\text{std}}^{n+1} + \beta_k \Phi_k^{n-1}}{1 + \beta_k}. \quad (20)$$

This is the form actually used in the implementation: no leapfrog kernel needs to be modified, only a single $\mathcal{O}(N_{\text{cells}})$ sweep per field per step. At $\beta_k = 0$ the formula reduces to $\Phi_{k,\text{std}}^{n+1}$ identically, preserving backward compatibility for tests that do not use the friction array.

4.4 Per-cell profile

The implementation stores $\beta_k = \gamma_k dt$ as a single floating-point array `friction_d[k]`. A practical profile is a small uniform floor plus a polynomial ramp to a larger value at the wall:

$$\beta_k = \begin{cases} \beta_{\text{floor}} & d_k \geq N_{\text{damp}} \\ \beta_{\text{floor}} + (\beta_{\text{wall}} - \beta_{\text{floor}}) u^2, \quad u = (N_{\text{damp}} - d_k)/N_{\text{damp}} & 0 \leq d_k < N_{\text{damp}} \end{cases} \quad (21)$$

with d_k the cell-count distance from the nearest wall. Sensible values for the web demo on a 384×384 grid with absorber depth $N_{\text{damp}} = 48$:

$$\beta_{\text{floor}} = 10^{-3}, \quad \beta_{\text{wall}} = 2 \times 10^{-2}. \quad (22)$$

Mode lifetimes: lowest standing mode ~ 1000 steps; modes localized within the ring ~ 50 steps. At $\text{CFL} = 1/\sqrt{2}$ this is well inside the stability region for both the bulk and the wall.

5 The DC mode under each boundary condition

The friction absorber damps every mode with nonzero $\partial_t \Phi$. The constant-everywhere mode $\Phi_{DC}(\mathbf{x}, t) = a(t)$ has $\partial_t a$ that is *not* zero in general – the wave equation drives it via the spatial mean of the source:

$$\ddot{a} = s_c \bar{\rho}, \quad \bar{\rho} = \frac{1}{|\Omega|} \int_{\Omega} \rho d^2x. \quad (23)$$

For a net-nonzero source $\bar{\rho} \neq 0$ (any scenario with nonzero total mass or charge), a undergoes constant acceleration: $a(t) = \frac{1}{2} s_c \bar{\rho} t^2$ plus initial conditions.

5.1 Dirichlet outer wall

Zero-Dirichlet at the outermost ring forces $a(t) \equiv 0$ at the wall, which under enforcement at every step pins the DC mode for the entire box. The friction term then sees no DC content, and the static fixed point is the discrete Dirichlet-Poisson solution. The *near-wall* value of Φ is forced to zero rather than continuing the L-W $\ln r$ tail; the L-W initialization must be tapered through the absorbing zone to match.

This is the configuration recommended for net-nonzero sources, and is what the v38 §9.6 absorber implicitly relied on (the multiplicative form does the pinning automatically).

5.2 Neumann outer wall

Zero-Neumann ($\partial_n \Phi = 0$ at the wall) leaves the DC mode unconstrained. With $\bar{\rho} \neq 0$ the DC mode runs away quadratically in time. For the demo’s `pic.binary` ($m_{\text{total}} = 0.02$, $WH = 147456$) at $\bar{\rho} \sim 10^{-7}$ and $s_c \sim 10$, the DC component grows past $|\Phi^{\text{static}}| \sim 10^{-1}$ in $\sim 10^3$ steps – visually rapid.

Two gauge-fix options:

1. **Zero-mean each step.** After the leapfrog rotation, subtract the spatial mean of Φ_g (and ϕ_{em}) from prev and curr. This is gauge-equivalent (Poisson is defined up to a constant; forces depend on gradients only). Tested implementation: works for `pic.static` but exhibits a small bulk-momentum drift in multi-particle scenarios – the per-step constant subtraction apparently interacts with the centered-friction term in a way that bleeds momentum out of the conserved sector. Cause not yet isolated; flagged for follow-up.
2. **Pin one cell.** Force Φ at a single fixed cell to zero each step. Equivalent to a one-point Dirichlet condition; preserves Neumann elsewhere. Not implemented in v42.

The pragmatic choice in v42 is to use **Dirichlet outer BC** plus friction. The friction does the work the multiplicative damping was supposed to do (absorbing the lowest mode), and the Dirichlet condition does the work the multiplicative damping was *also* doing as a side effect (pinning the DC). Separation of concerns.

6 Summary of replacement algorithms

Replacement for v38 §15.9 (field initialization)

Place all particles at their scenario IC positions and momenta.

Evaluate Eqs. (??)-(??) at every grid cell for every particle. Clip r at $\frac{1}{2}dx$ to avoid the log singularity at the source.

If the outer BC is Dirichlet, multiply through by the polynomial- $m = 2$ taper that brings the field smoothly to zero at the wall over the depth of the absorbing zone. If Neumann, do not taper – the discontinuity would launch an inward wave on step 1.

Copy the resulting `curr` buffer into `prev` and `next` so the encoded time derivative is identically zero.

Begin the wave-equation evolution. No half-step momentum back-extrapolation is needed because the field is already at the static fixed point.

Replacement for v38 §9.6 (absorbing boundary layer)

Allocate `friction_d[k]` of size $W \times H$.

Populate with a tapered profile: β_{floor} in the interior, smoothly increasing to β_{wall} over the last N_{damp} cells from each wall.

In `gr_sim_step`, after the standard leapfrog computes `next` for every field but *before* the three-pointer rotation, apply the per-cell correction (??).

Outer-wall boundary: zero-Dirichlet (`next[outer ring] = 0`) is recommended for net-nonzero sources. Neumann is available as a flag but requires an additional gauge fix; see §??.

No multiplicative damping is applied – the entire absorber function is carried by the friction.

7 Status & open items

What works. The friction absorber + Dirichlet + L-W initialization combination reaches a stable post-transient configuration on both `pic_static` (single mass) and `pic_binary` (two orbiting masses):

- Initial Dirichlet-induced inward wave from the L-W taper at the wall damps within a few hundred steps.
- No lowest-mode trap; no inversion oscillation.
- The bound binary keeps orbiting after the transient.
- Discrete-Dirichlet-Poisson static is reached and held.

Open follow-ups. Tracked separately:

- *COM drift on pic_binary.* After the boundary transient damps, the two-body center of mass moves at small constant velocity. Suspected cause: the transient imparts a momentum impulse to the pair that is not exactly equal and opposite due to sub-cell grid alignment of the particles. Fix is to make the L-W IC consistent with the discrete Dirichlet equilibrium so no transient is launched, either via TSC-convolved L-W or via a few SOR sweeps applied to the L-W as a smoother.
- *Continuum-discrete Poisson residual.* See §??. Currently dissipated by friction; could be eliminated by TSC-convolved L-W.
- *Neumann + zero-mean momentum bleed.* See §??, item 1. Currently worked around by using Dirichlet.